Predicting Emotional Valence and Arousal Over Time: A Hybrid Approach Combining Lexical and Contextual Representations

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Abstract

How can lexical, contextual, temporal and user-level information be combined to predict both within-person emotional fluctuations and between-person emotional tendencies in longitudinal ecological momentary assessment (EMA) texts? We study this in a controlled ablation of eleven models: feature groups are added one at a time to a TF-IDF Ridge regression model and to a fine-tuned multilingual DistilBERT model, and the strongest model of each family is combined in a weighted ensemble. On the test set, the ensemble reaches the highest composite Pearson correlation for valence (0.66), and TF-IDF with user baseline features is best for arousal (0.51); bootstrap confidence intervals show that these two systems are not reliably different from each other. The two model families have complementary strengths: on the test set, DistilBERT tracks within-person fluctuations better, while TF-IDF better separates users with different emotional baselines, and the ensemble obtains the highest within-person correlation for both dimensions. We also find that user baseline features help on the test set, where 40% of entries come from users seen during training, but hurt on our user-disjoint validation split, so the evaluation design largely decides whether personalisation appears useful.

1 Introduction

Understanding how human emotions fluctuate over time is a central problem in affective computing, with applications ranging from mental health monitoring to personalised dialogue systems. Traditional sentiment analysis reduces emotional experience to a single positive–negative axis (Pang and Lee, 2008). The circumplex model of affect (Russell, 1980) instead describes emotional states in two dimensions: valence (pleasantness) and arousal (activation). A person can feel calm and content (positive valence, low arousal) or anxious and agitated (negative valence, high arousal), a distinction a single axis cannot capture.

Emotion prediction is particularly challenging in ecological momentary assessment (EMA), where participants repeatedly report their emotional state over long periods through short self-written texts (Shiffman et al., 2008). This longitudinal structure creates two prediction objectives. First, a model must capture *within-person* variation: how does an individual's state change from entry to entry? Second, it must capture *between-person* differences: are some people generally more positive or more energetic than others? Within-person prediction rewards sensitivity to the content of each entry, while between-person prediction rewards stable estimates of each person's baseline.

This paper presents our system for WNNLP 2026 Track 3, which uses the data and official evaluation of SemEval-2026 Task 2 (Subtask 1). The primary metric is a composite Pearson correlation that rewards both objectives at once. Our contributions are:

- A controlled ablation over eleven configurations that isolates the contribution of structural, temporal and user-level features and of text-type specialisation, for both a lexical and a contextual model family.
- Evidence that the two families have complementary strengths on the test set, with DistilBERT stronger within persons and TF-IDF stronger between persons, and a weighted ensemble that obtains the best within-person correlation and the best valence composite.
- An analysis showing that the apparent value of user baseline features depends on whether evaluation users were seen in training, together with an error analysis of the strongest single model.

2 Related Work

Dimensional emotion prediction. Predicting continuous affective dimensions from text builds on the dimensional models of Russell (1980) and Mehrabian (1996). Strapparava and Mihalcea (2007) introduced the SemEval-2007 Affective Text task on news headlines. Buechel and Hahn (2017) presented EmoBank, a corpus annotated for valence, arousal and dominance. Preoţiuc-Pietro et al. (2016) annotated Facebook posts for valence and arousal and built models that predict both dimensions from text. We address the harder setting of repeated, person-specific reports over time.

EMA and language. EMA is widely used in psychology and clinical research to capture emotional states in natural settings (Shiffman et al., 2008), and language data play a growing role in computational mental health research (Coppersmith et al., 2015). Park et al. (2015) showed that the language people write on social media predicts their self-reported personality.

User-level modelling. Word meaning and sentiment vary across communities and domains (Hamilton et al., 2016), and modelling demographic language variation improves sentiment classification (Volkova et al., 2013). Lynn et al. (2020) showed that aggregating a user's messages hierarchically with message-level attention improves user-level prediction. These results motivate our user baseline features.

Transformers and ensembles. Pretrained transformers (Devlin et al., 2019) are the standard choice for tasks that require contextual understanding; on TweetEval, which includes emotion and sentiment classification, fine-tuned transformers outperform traditional baselines (Barbieri et al., 2020). We use multilingual DistilBERT (Sanh et al., 2019), a distilled variant of BERT that keeps most of its performance with 40% fewer parameters. Averaging models with different inductive biases is most effective when their errors are weakly correlated (Dietterich, 2000), which motivates our hybrid model.

3 Task and Data

3.1 Task definition

Given a text entry and its metadata (timestamp, collection phase, text type), the task is to predict two ordinal self-reports: valence $v \in \{-2, \ldots, 2\}$ and arousal $a \in \{0, 1, 2\}$. Because the official evaluation uses Pearson correlation, we treat both as regression targets and predict real values. The two dimensions are evaluated separately.

Split	Texts	Users	Essays	Word lists
Train	2,764	137	1,331	1,433
Test	1,737	91	807	930

Table 1: Dataset statistics.

3.2 Dataset

The data are EMA entries collected between 2021 and 2024 (Table 1). Participants contributed on average 20.2 training entries across up to seven collection phases. 46 of the 91 test users also appear in the training data, and their entries make up 40% of the test set.

There are two text types. Essays are free-form paragraphs averaging 58.8 words ($SD = 27.4$). Feeling-word lists are comma-separated emotion words averaging 8.9 tokens ($SD = 1.8$). Valence in the training data is 32% neutral, 41% positive and 27% negative. Arousal is skewed towards low values: 44% of entries have arousal 0, 37% arousal 1 and 19% arousal 2.

3.3 Evaluation metrics

We use the three Pearson correlations of the official evaluation, computed separately for valence and arousal. The **within-person correlation** r_{within} is the mean of per-user correlations between predictions and gold labels. The **between-person correlation** r_{between} is the correlation between per-user mean predictions and per-user mean labels. The **composite correlation**, our primary metric, combines both with Fisher's z-transformation:

$$r_{\text{comp}} = \tanh\left(\tfrac{1}{2}\big(z(r_{\text{within}}) + z(r_{\text{between}})\big)\right), \quad (1)$$

where $z(r) = \tanh^{-1}(r)$. We also report the official composite MAE (lower is better), which combines per-user and between-user MAE in the same way.

To judge whether differences between models are meaningful, we compute 95% confidence intervals for test composites with a bootstrap over users (2,000 resamples).

4 System Description

We organise the experiments as a controlled ablation. Feature groups are added in the same order to

a lexical and a contextual model, which separates the effect of the model family from the effect of the features. All models predict valence and arousal jointly with independent outputs.

Validation. We hold out 15% of the training users with a grouped split, so no user appears in both the training and validation parts (512 entries from 21 users). The text-type models M5 and M10 are trained separately for essays and word lists, and the split is made separately for each text type. Their validation set therefore differs (571 entries from 32 users), and a user can appear in training for one text type and in validation for the other.

4.1 Feature groups

The **text representation** is either TF-IDF over unigrams and bigrams or the final-layer [CLS] embedding of multilingual DistilBERT.

Structural and intensity features describe surface form: character length, word count, text type, counts of exclamation marks, question marks and uppercase letters, and the uppercase ratio.

Temporal features encode the longitudinal design: the collection phase and two time gaps, days since the user's first entry and days since their previous entry. Because of a timestamp parsing error in the runs reported here, the two time gaps were zero for every entry, so this group effectively contains only the collection phase. Rerunning the TF-IDF models with correct parsing changes their composite scores by at most 0.025 and does not change any conclusion.

User baseline features are the user's mean valence and mean arousal in the training data. Users without training entries receive the global training mean. For validation, these means are computed from the training part only. For test predictions, M4 and M5 use all training data, while M9 and M10 use the training part of the split.

All numeric features are standardised with statistics from the training data only.

4.2 Model configurations

Table 2 lists the eleven configurations. The TF-IDF models use unigrams and bigrams (`min_df`=2, at most 20,000 features) and Ridge regression ($\alpha = 1.0$), and are retrained on all training data for the test predictions.

The DistilBERT models use `distilbert-base-multilingual-cased`. The [CLS] embedding is concatenated with the numeric features and passed to a two-layer head (128 hidden units, ReLU, dropout 0.25), following common practice of fine-tuning on the [CLS] representation (Sun et al., 2019). We train with AdamW (learning rate 10^{-5}, weight decay 0.01, batch size 16, maximum length 128) for up to 8 epochs with linear warm-up over the first 10% of steps, and keep the epoch with the lowest validation loss. Because the validation part is needed for this model selection, the DistilBERT models are trained on the training part only (85% of users). We did not compare the multilingual model with the English-only DistilBERT, which might suit this mostly English data better. All experiments use random seed 42.

ID	Configuration
M1	TF-IDF, text only
M2	+ structural/intensity
M3	+ temporal
M4	+ user baselines
M5	M4 with text-type split
M6	DistilBERT, text only
M7	+ structural/intensity
M8	+ temporal
M9	+ user baselines
M10	M9 with text-type split
M11	Weighted ensemble of M5 and M10

Table 2: Model configurations. Within each family, feature groups are added cumulatively.

4.3 Hybrid ensemble

The final system averages the predictions of M5 and M10:

$$\hat{y}_{\text{hybrid}} = w\,\hat{y}_{\text{TF-IDF}} + (1 - w)\,\hat{y}_{\text{DistilBERT}}. \quad (2)$$

We choose w from $\{0.0, 0.1, \ldots, 1.0\}$ on the shared validation set of M5 and M10 by maximising the mean of the pooled Pearson correlations for valence and arousal. This selects $w = 0.4$. Maximising the mean composite correlation instead would select $w = 0.3$.

5 Results

Table 3 shows the test results, and Table 4 the validation results.

Overall. On the test set, the hybrid ensemble M11 has the highest valence composite (0.658, 95% CI 0.58–0.73), and M4 the highest arousal composite (0.509, CI 0.41–0.60). Neither difference between the two is reliable: the bootstrap

Model	Valence				Arousal			
	r_{between}	r_{within}	r_{comp}	MAE_{comp}	r_{between}	r_{within}	r_{comp}	MAE_{comp}
M1: TF-IDF text only	0.714	0.448	0.597	0.681	0.564	0.311	0.447	0.414
M2: + structural/intensity	0.685	0.454	0.581	0.687	0.563	0.340	0.458	0.410
M3: + temporal	0.707	0.457	0.596	0.674	0.555	0.340	0.454	0.411
M4: + user baselines	**0.774**	0.439	0.635	0.640	**0.640**	0.350	**0.509**	**0.383**
M5: + text-type split	0.733	0.442	0.608	0.657	0.588	0.295	0.454	0.400
M6: DistilBERT text only	0.674	0.535	0.609	0.677	0.382	0.377	0.379	0.434
M7: + structural/intensity	0.690	0.535	0.618	0.691	0.529	0.378	0.457	0.415
M8: + temporal	0.715	0.531	0.632	0.651	0.454	0.361	0.409	0.422
M9: + user baselines	0.664	0.491	0.584	0.684	0.449	0.333	0.392	0.434
M10: + text-type split	0.688	0.538	0.619	0.679	0.336	0.361	0.349	0.433
M11: Hybrid (M5 + M10)	0.748	**0.546**	**0.658**	**0.620**	0.586	**0.389**	0.494	0.396

Table 3: Test results (1,737 entries, 91 users). r_{comp} is the primary metric; MAE_{comp} is lower-is-better. Best value per column in bold.

Model	Valence				Arousal			
	r_{between}	r_{within}	r_{comp}	MAE_{comp}	r_{between}	r_{within}	r_{comp}	MAE_{comp}
M1: TF-IDF text only	0.642	0.504	0.577	0.763	0.139	0.234	0.187	0.428
M2: + structural/intensity	0.626	0.500	0.566	0.764	0.160	0.241	0.201	0.428
M3: + temporal	0.632	0.498	0.569	0.761	0.162	0.242	0.202	0.425
M4: + user baselines	0.624	0.441	0.539	0.767	−0.019	0.312	0.151	0.431
M5†: + text-type split	0.532	0.453	0.493	0.926	0.360	0.295	0.328	0.404
M6: DistilBERT text only	0.725	0.612	0.672	0.684	0.201	0.272	0.236	0.414
M7: + structural/intensity	0.740	0.629	**0.688**	0.680	0.133	0.292	0.214	0.426
M8: + temporal	0.719	0.630	0.677	**0.661**	0.350	0.277	0.314	0.400
M9: + user baselines	**0.743**	0.518	0.644	0.689	0.082	0.320	0.204	0.425
M10†: + text-type split	0.564	**0.647**	0.607	0.764	0.456	0.345	0.402	0.397
M11†: Hybrid (M5 + M10)	0.574	0.645	0.610	0.751	**0.470**	**0.354**	**0.413**	**0.387**

Table 4: Validation results. Models without † are evaluated on 512 entries from 21 users; models with † on a different split of 571 entries from 32 users (Section 4), so the two groups are not directly comparable. Best value per column in bold.

interval of the difference includes zero for both valence (−0.04 to 0.08) and arousal (−0.04 to 0.08). The ensemble is, however, reliably better than both of its components for valence (by 0.05 over M5 and 0.04 over M10, both intervals above zero) and better than M10 for arousal (by 0.15). Most other differences between neighbouring configurations are within the bootstrap uncertainty, so we focus on consistent patterns rather than single rankings.

TF-IDF progression. Structural and temporal features change the test composite by at most 0.016. Adding user baselines (M4) gives the largest gain in the TF-IDF family, mainly through higher between-person correlation (valence 0.707 → 0.774, arousal 0.555 → 0.640). Splitting by text type (M5) lowers both composites.

DistilBERT progression. DistilBERT models have higher within-person correlations than all TF-IDF models for valence (0.49–0.54 vs. 0.44–0.46). Structural features improve arousal (M6 → M7: 0.379 → 0.457). Unlike for TF-IDF, user baselines lower the test composite (M8 → M9), significantly so for valence (difference −0.05, CI −0.08 to −0.01). Part of the reason is likely that M9 computes baselines from only 85% of the training users and selects its epoch on validation users for whom the baseline carries no information.

Validation versus test. Validation and test results disagree in several places. The clearest case is arousal, where the validation composites (0.15–0.41) are far below the test composites (0.35–0.51). The validation set contains only 21 or 32 users, so between-person correlations computed over so few user means are very noisy, and model choices made on validation should be read with this in mind.

6 Analysis

6.1 Within-person versus between-person trade-off

Figure 1 plots r_{within} against r_{between} on the test set. The two families occupy different regions. For valence, every TF-IDF model has lower within-person correlation than every DistilBERT model, while TF-IDF models tend to have higher between-person correlation. For arousal the separation in r_{between} is larger: TF-IDF models reach 0.56–0.64, DistilBERT models 0.34–0.53. The ensemble obtains the highest within-person correlation for both dimensions (0.546 and 0.389) while keeping a between-person correlation close to the TF-IDF models, which is why it has the best valence composite.

A plausible explanation is that TF-IDF features are dominated by frequent emotion words whose use is fairly stable within a person, which helps separate people, whereas DistilBERT is more sensitive to the phrasing of each entry, which helps track change over time. On the validation set the within-person pattern is the same, but the between-person pattern does not hold for valence, where DistilBERT models are higher; with 21 validation users we do not consider this conclusive either way.

6.2 User baselines depend on the evaluation design

User baselines lower the validation composite in both families (M3 $\rightarrow$ M4, M8 $\rightarrow$ M9), most strikingly for arousal in M4, where r_{between} falls to -0.02. On the test set, the same features give the best TF-IDF model. The difference follows from how the splits are built. In our validation split, every user is new, so the baseline feature equals the global mean for all validation entries and carries no user information. On the test set, 40% of entries come from users who appear in training, and for them the baseline is informative.

The negative validation result for M4 is also plausible: during training the user mean explains much of the between-person variance, partly because it is computed from the same labels the model is trained on, so the model learns to rely on it rather than on the text for user-level differences. When the feature becomes constant, the text-based between-person signal is weaker than in M3. Computing the training baselines out of fold would avoid this.

The practical lesson is that the value of personalisation features depends on whether the evaluation contains known users. A user-disjoint validation split, which is otherwise good practice, underestimates them when the test set contains known users.

Type	n	r_{val}	r_{aro}	MAE_{val}	MAE_{aro}
Essay	239	0.574	0.342	0.884	0.542
Word list	332	0.663	0.604	0.798	0.566

Table 5: M10 validation error by text type (pooled Pearson r and MAE).

6.3 Text-type specialisation

Training separate models for essays and word lists does not help consistently. On the test set it lowers the TF-IDF composites (M4 $\rightarrow$ M5) and, for DistilBERT, improves valence slightly but lowers arousal (M9 $\rightarrow$ M10). On validation, M5 and M10 appear to improve arousal, but they are evaluated on a different validation set from the other models, so this comparison is not reliable.

6.4 Why arousal is harder than valence

On the test set, arousal composites are 0.12–0.27 lower than valence composites for every model. Three factors likely contribute: the arousal scale is shorter, valence is more directly expressed by individual words (Liu, 2012), and the arousal distribution is skewed, with 44% of entries at 0 and only 19% at 2. Annotation studies likewise report lower agreement for arousal than for valence (Buechel and Hahn, 2017).

6.5 Error analysis

We analyse the validation predictions of M10 (571 entries, 32 users), the DistilBERT model with the highest within-person correlation.

Text type. Word lists are predicted more accurately than essays, especially for arousal (Table 5). Word lists name emotional states directly (e.g. *exhausted, anxious, content*), while essays require inferring emotion from narrative that often contains hedging or mixed signals.

True label value. The model strongly regresses towards the mean (Table 6, Figure 2). For entries with valence -2, the mean prediction is -0.60, close to neutral. For arousal, entries labelled 0 are over-predicted (mean prediction 0.61) and entries labelled 2 are under-predicted (mean prediction 1.05).

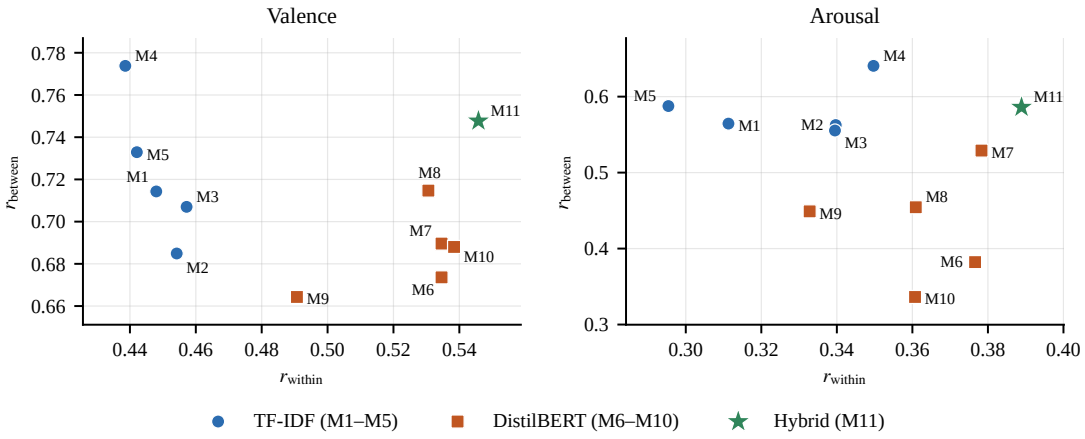


Figure 1: Within-person vs. between-person correlation on the test set for all eleven models. The ideal model lies in the top-right corner.

True	n	MAE	Mean pred.	Bias
Valence				
-2	72	1.399	-0.601	$+1.399$
-1	74	0.596	-0.496	$+0.504$
0	151	0.857	$+0.493$	$+0.493$
$+1$	126	0.579	$+0.743$	-0.257
$+2$	148	0.871	$+1.129$	-0.871
Arousal				
0	290	0.608	$+0.608$	$+0.608$
1	181	0.254	$+0.886$	-0.114
2	100	0.954	$+1.046$	-0.954

Table 6: M10 validation error by true label. Bias = mean prediction − true label.

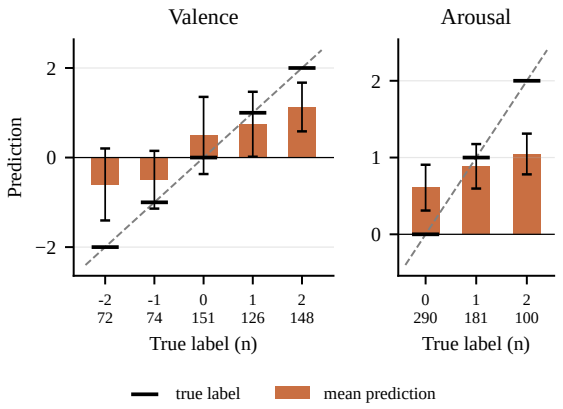


Figure 2: Mean M10 validation prediction for each true label (bars, with ±1 standard deviation). Black ticks mark the true label, the dashed line marks perfect prediction, and the numbers below each label give the number of entries.

Qualitative examples. The largest errors involve positive surface language with a negative self-report. The entry *"I am happy to have had a wonderful, delicious lunch with my husband. It made him happy and content…"* is labelled valence -2 but predicted $+1.44$; the positive words dominate while the underlying state is missed. Similarly, *"I am feeling better than yesterday and have a better mood as well. I am still low on my self esteem…"* is labelled -2 and predicted $+1.24$. The best predictions are for direct statements, such as *"Right now, my mood is neutral. Nothing too crazy is happening…"* (true $v = 0$, $a = 1$; predicted $v = -0.01$, $a = 1.06$).

Users seen under the other text type. Because the M10 split is made separately per text type, 200 validation entries come from users whose entries of the other type are in the training part. These entries are predicted worse than entries from entirely new users (valence r 0.459 vs. 0.746, MAE 0.989 vs. 0.750; arousal r 0.485 vs. 0.511). Since user baselines are computed per text type, both groups receive the global mean as baseline, so this gap reflects differences between the users in the two groups rather than an effect of the baseline feature.

Collection phase. Valence MAE is highest in phases 1 and 7 (0.964 and 0.937) and lowest in phase 4 (0.656), possibly reflecting greater emotional variability at the start and end of the study.

7 Conclusion

We compared eleven configurations for predicting valence and arousal from longitudinal EMA texts. On the test set, fine-tuned DistilBERT tracked within-person fluctuations better, TF-IDF better separated users with different emotional baselines,

and a weighted ensemble of the two obtained the highest within-person correlation for both dimensions and the best valence composite (0.66). For arousal, TF-IDF with user baselines was best (0.51), and the ensemble was statistically indistinguishable from it. We also found that user baseline features help when test users have been seen in training but hurt on a user-disjoint validation split, which shows that the evaluation design largely decides whether personalisation appears useful. Future work should use validation splits that mirror the test distribution of seen and unseen users, compute user features out of fold, and explore models that adapt to both kinds of variation.

Limitations

All results come from a single run with one random seed, and the validation set contains only 21 or 32 users, so model and ensemble-weight selection are noisy. Because of a timestamp parsing error, the two time-gap features were always zero in the reported runs; correcting it changes TF-IDF composites by at most 0.025, but the DistilBERT models have not been rerun. User baselines for training entries include the entry's own label, which may encourage over-reliance on them. The DistilBERT models were trained on 85% of the training users, so they saw less data than the TF-IDF models. The ensemble weight was selected with pooled rather than composite correlation. Finally, we did not compare multilingual and English-only DistilBERT or try domain-adaptive pretraining.

Code Availability

Code, predictions and evaluation scripts are available at https://github.com/S2k22/Predicting-Emotional-Valence-and-Arousal-Over-Time.

AI Assistance Disclosure

The authors used AI assistance (UiO GPT) for code development support and manuscript writing. All research design, model development, experimental execution and scientific interpretation are the authors' own.

Revision Note

This version corrects the version reviewed for WNNLP 2026. The test table previously repeated the validation results and has been replaced with the actual test results, which changed several conclusions. We also corrected the reported DistilBERT hyperparameters and ensemble selection criterion, documented the timestamp parsing error, recomputed the seen-user analysis, added bootstrap confidence intervals, revised Figure 2, and corrected several references.

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